

POTAPOVO, Russian Federation

WMO: 231740

Lat: 68.681N

Lon: 86.273E

Elev: 135

StdP: 14.62

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-54.8	-50.2	-58.3	0.2	-54.5	-54.0	0.2	-50.0	40.7	-8.9	35.9	-11.9	1.2	180	0.844

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	78.2	63.3	73.2	61.8	69.1	59.6	65.5	74.3	63.2	71.0	60.7	67.7	6.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	83.9	68.3	59.8	77.2	67.4	57.1	69.8	64.5	30.5	73.6	28.7	71.2	27.0	68.1	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.1	21.8	19.0	DB	-55.4	81.4	4.2	6.1	-58.4	85.8	-60.9	89.4	-63.2	92.8	-66.3	97.2	
			WB	-56.2	67.7	4.9	4.4	-59.8	70.8	-62.7	73.4	-65.5	75.9	-69.1	79.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	-19.7	-13.7	-0.5	10.0	27.5	48.6	58.1	51.1	40.6	20.3	-6.8	-10.1	(6)
(7)		DBStd	29.92	18.61	17.02	17.32	14.35	9.48	11.01	8.13	6.67	8.10	12.18	16.04	18.56	(7)
(8)		HDD50	12439	2162	1783	1567	1199	699	158	20	66	298	921	1703	1863	(8)
(9)		HDD65	17451	2627	2203	2032	1649	1164	501	244	431	733	1386	2153	2328	(9)
(10)		CDD50	504	0	0	0	0	0	117	271	101	15	0	0	0	(10)
(11)		CDD65	42	0	0	0	0	0	10	30	2	0	0	0	0	(11)
(12)		CDH74	324	0	0	0	0	0	76	246	2	0	0	0	0	(12)
(13)		CDH80	50	0	0	0	0	0	10	40	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.8	8.2	8.7	7.4	7.6	7.7	7.6	6.9	6.9	7.6	8.4	7.6	8.8	(14)
(15)	Precipitation	PrecAvg	34.3	2.8	2.5	2.4	2.4	2.4	2.3	2.4	3.1	2.7	4.3	3.3	3.4	(15)
(16)		PrecMax	43.7	5.7	4.7	5.2	4.8	4.1	6.3	5.3	5.6	7.0	6.4	5.6	(16)	
(17)		PrecMin	17.0	0.5	0.5	0.4	0.1	0.5	0.3	0.4	1.1	1.2	1.3	0.7	0.3	(17)
(18)		PrecStd	5.5	1.3	1.1	1.2	1.1	0.9	1.0	1.3	0.9	1.0	1.5	1.4	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	22.5	33.3	44.9	54.2	81.8	84.0	73.2	65.0	46.2	29.7	24.7	(19)
(20)			MCWB	18.2	21.5	31.6	37.0	44.6	65.2	65.6	63.1	54.4	44.6	28.3	23.6	(20)
(21)		2%	DB	13.3	19.1	30.7	38.7	48.5	76.1	80.6	68.9	59.1	40.7	26.1	21.2	(21)
(22)			MCWB	12.4	18.4	28.9	34.2	41.6	60.9	65.6	60.3	52.6	39.4	25.1	20.1	(22)
(23)		5%	DB	9.4	14.0	28.1	33.8	43.2	70.7	76.5	65.4	55.6	37.4	22.5	18.6	(23)
(24)			MCWB	8.6	13.2	26.8	30.9	38.2	58.5	63.2	58.3	50.4	35.9	21.6	17.7	(24)
(25)		10%	DB	5.3	9.5	24.5	30.7	39.6	65.6	72.0	61.9	51.8	34.1	16.6	14.8	(25)
(26)			MCWB	4.5	8.8	23.2	28.5	35.4	56.3	61.7	56.6	47.7	32.5	15.7	13.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	21.9	31.9	38.1	45.2	65.6	69.9	65.5	57.2	44.9	29.1	23.7	(27)
(28)			MCD B	18.4	22.5	33.0	43.4	53.5	76.8	78.8	71.4	61.4	46.9	30.0	24.4	(28)
(29)		2%	WB	12.4	18.4	29.2	34.3	41.6	62.5	66.7	62.1	53.8	39.8	25.5	20.2	(29)
(30)			MCD B	12.9	18.9	30.3	38.5	47.7	72.1	76.4	66.7	58.4	40.8	26.3	21.0	(30)
(31)		5%	WB	8.5	13.3	26.6	31.1	38.7	59.9	64.8	59.4	50.8	36.0	21.5	17.9	(31)
(32)			MCD B	9.2	13.9	28.1	33.5	43.0	68.8	73.3	63.8	55.2	37.0	22.4	18.7	(32)
(33)		10%	WB	4.5	8.7	23.4	28.6	35.8	57.0	62.9	57.0	48.0	33.1	15.7	14.1	(33)
(34)			MCD B	5.2	9.5	24.5	30.9	38.8	65.5	70.6	61.5	50.8	34.1	16.6	15.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.8	13.9	17.4	17.1	11.3	13.6	15.0	11.6	9.6	9.6	13.6	14.1	(35)
(36)			MCDBR	15.7	15.8	14.0	17.0	14.3	20.2	21.7	16.6	13.8	8.1	14.9	14.0	(36)
(37)			MCWBR	15.4	15.1	13.3	14.5	9.3	10.7	10.8	10.0	9.0	7.4	15.0	13.3	(37)
(38)		5% WB	MCD BR	15.6	14.4	14.7	16.4	13.5	18.8	18.0	13.6	11.9	7.7	16.1	13.8	(38)
(39)			MCWBR	15.4	14.2	14.1	14.6	9.0	11.2	10.7	9.5	8.8	7.2	15.8	13.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.216	0.253	0.305	0.328	0.354	0.369	0.404	0.393	0.328	0.255	0.199	N/A	(40)
(41)		taud		1.857	1.971	2.005	2.050	2.123	2.378	2.285	2.272	2.407	2.109	1.885	N/A	(41)
(42)		Ebn at Noon		12	154	218	253	261	263	246	232	222	155	12	0	(42)
(43)		Edh at Noon		2	16	29	38	40	32	34	31	21	13	2	0	(43)
(44)	All-Sky Solar Radiation	RadAvg		6	137	627	1308	1789	1716	1598	993	524	171	23	0	(44)
(45)		RadStd		1	16	68	106	143	155	210	115	55	31	3	0	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page